

Ch 5: Minerals and Energy Resources

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What is a mineral? Give two examples of minerals found in India.

Q2. What is an 'ore'? How does it differ from a mineral?

Q3. How do minerals occur in igneous and metamorphic rocks? Give two examples. (PYQ 2016)

Q4. How do minerals occur in sedimentary rocks? Give two examples. (PYQ 2015)

Q5. What are 'placer deposits'? Give two examples of minerals found in them.

Q6. What are 'veins and lodes'? Give two examples of minerals found in them.

Q7. Distinguish between metallic and non-metallic minerals with two examples each. (PYQ 2013)

Q8. Distinguish between ferrous and non-ferrous minerals with two examples each. (PYQ 2013, 2017)

Q9. What is Magnetite? What makes it the finest iron ore?

Q10. What is Haematite? Why is it the most important industrial iron ore?

Q11. What is the use of manganese? Name the largest manganese-producing state of India.

Q12. How much manganese is required to manufacture one tonne of steel?

Q13. What is bauxite? From which mineral is aluminium extracted? Name the largest bauxite-producing state.

Q14. Why is aluminium gaining importance as a metal? Give two reasons.

Q15. What are the properties of mica that make it indispensable in electrical and electronic industries?

Q16. Where are mica deposits mainly found in India? Name two states.

Q17. What is limestone? Name two industries that depend on it as a raw material.

Q18. What are conventional sources of energy? Give three examples. (PYQ 2012)

Q19. What are non-conventional sources of energy? Give three examples. (PYQ 2012)

Q20. What is coal? Name the four types of coal found in India in order of carbon content.

Q21. Which is the most abundantly found coal in India? Where is it mainly found?

Q22. Where is Digboi oil field located? Why is it historically significant?

Q23. What is natural gas? Why is it considered a clean source of energy? (PYQ SQP 2017)

Q24. What is biogas? What raw materials are used to produce it? (PYQ)

Q25. What is solar energy? Name two states in India with high solar energy potential.

Q26. What is tidal energy? Name one location in India where it can be harnessed.

Q27. What is geothermal energy? Name two geothermal energy projects in India.

Q28. What is nuclear energy? Name two minerals used as nuclear fuel in India.

Q29. Where are monazite sands found in India? What mineral do they contain?

Q30. Name two major coalfields in India and the states where they are located.

Q31. Why are minerals described as non-renewable resources? Why must they be conserved?

Q32. Name the Odisha-Jharkhand iron ore belt. Name two mines in each state.

SECTION B - 3 MARKS

Q1. Explain the three main modes of occurrence of minerals in India with examples. (PYQ 2016)

Q2. Distinguish between ferrous and non-ferrous minerals. Give two examples of each and explain their uses. (PYQ 2013)

Q3. Describe any three characteristics of ferrous minerals found in India. (PYQ 2011)

Q4. Explain the importance of iron ore in India's industrial development. Name the four major iron ore belts.

Q5. Describe any three characteristics of the Odisha-Jharkhand iron ore belt. (PYQ 2017)

Q6. Explain the importance of manganese as a raw material. What are its uses? Name the largest producing state. (PYQ)

Q7. Describe the formation, importance, and distribution of bauxite in India. (PYQ)

Q8. Explain the properties and uses of mica. Why is it called the most indispensable mineral for electrical industries?

Q9. Explain the importance of copper as a non-ferrous mineral. Where is it mainly found in India?

Q10. Distinguish between metallic and non-metallic minerals. Give three examples of each and their uses. (PYQ 2013)

Q11. Explain the classification of minerals on the basis of chemical and physical properties. Give examples.

Q12. Explain three factors that affect the economic viability of a mineral reserve.

Q13. Explain why minerals need to be conserved. Describe three measures for mineral conservation.

Q14. Distinguish between conventional and non-conventional sources of energy with three examples each. (PYQ 2013, 2014)

Q15. Explain the importance of coal as an energy resource in India. Name the main types of coal and their uses.

Q16. Explain the distribution of petroleum resources in India. Name the three major oil-producing regions.

Q17. "Natural gas is an important source of clean energy." Support with three examples. (PYQ SQP 2017)

Q18. What is biogas? How is it produced? Explain its twin benefits for farmers. (PYQ)

Q19. Explain the advantages of solar energy as a non-conventional source. How can it be harnessed in India?

Q20. Explain the potential of wind energy in India. Name two states with the highest wind energy generation.

Q21. Explain what nuclear energy is. What minerals are used as fuel? Name two nuclear power stations in India.

Q22. Explain the advantages of non-conventional energy sources over conventional energy sources. Give three points.

Q23. "Energy is an indispensable requirement of modern life." Explain with three examples. (PYQ 2011)

Q24. Explain any three steps to conserve energy resources in India. (PYQ 2011)

Q25. Explain the distribution of mica in India. Which region is the most important? Why is Jharkhand significant?

Q26. Explain three environmental problems caused by mining. How can they be minimised?

Q27. Explain the formation and distribution of coal in India. Why are Gondwana coalfields more important?

Q28. Explain the uses and distribution of limestone in India. Why is it important for industry?

Q29. Explain tidal energy and geothermal energy as sources of non-conventional energy. Give examples of locations in India.

Q30. How is petroleum formed? In which rock structures does it occur? Explain with an example.

Q31. Explain the role of the Hazira-Vijaypur-Jagdishpur (HVJ) pipeline in India's natural gas distribution.

Q32. Describe any three characteristics of the Bellary-Chitradurga-Chikmagalur-Tumkur belt of iron ore.

Q33. Explain why India, despite having rich mineral resources, is not able to perform to its full potential in iron and steel production. (PYQ 2016)

Q34. Explain the importance of recycling minerals. How does it help in conservation and sustainability?

Q35. Why is the Chota Nagpur Plateau region called the mineral heartland of India? Explain three reasons.

Q36. Explain three challenges faced by the conventional energy sector in India and how non-conventional energy can address them.

Q37. Describe the uses of petroleum and its by-products in different industries. Give three examples.

Q38. Explain the potential and present status of solar energy development in India with examples of states and projects.

Q39. Explain the significance of tidal energy in India. Why is the Gulf of Khambhat considered ideal for tidal power?

Q40. Describe the properties of Magnetite and Haematite iron ore. Compare their iron content and industrial uses.

Q41. Explain three ways in which over-extraction of minerals harms the environment and local communities.

Q42. Describe the distribution of coal resources in India across the Damodar Valley, Godavari Valley, and Mahanadi basin.

SECTION C - 5 MARKS

Q1. Describe the geographical conditions, distribution, and importance of rice cultivation in India. (*PYQ — frequently repeated*)

Q2. Describe the geographical conditions, distribution, and importance of wheat cultivation in India.

Q3. Describe the geographical conditions, distribution, and importance of cotton cultivation in India. How is it linked to the textile industry?

Q4. Describe the geographical conditions, distribution, and importance of sugarcane cultivation in India. Which state leads and why?

Q5. Describe the geographical conditions, distribution, and importance of jute cultivation in India. Why is it losing market share?

Q6. Explain the types of farming practised in India. Describe primitive subsistence, intensive subsistence, commercial, and plantation farming with examples. (*PYQ — most repeated*)

Q7. "India is an agrarian country." Justify this statement by explaining the role of agriculture in India's economy with five points.

Q8. Explain the three cropping seasons of India — Kharif, Rabi, and Zaid — with sowing time, harvesting time, climate needed, and two crops each. *(PYQ — very frequently asked)*

Q9. Explain the impact of globalisation on Indian agriculture. What challenges has it created for small farmers?

Q10. "The growth rate in agriculture is decelerating." Explain the causes and describe the steps taken by the government to modernise agriculture. *(PYQ)*

Q11. Describe the Green Revolution — its causes, methods, areas where it succeeded, positive effects, and negative consequences. *(PYQ — frequently repeated)*

Q12. Explain any five factors responsible for low productivity of crops in India. What measures can improve agricultural productivity?

Q13. Describe the features of plantation agriculture in India. Explain with reference to tea, coffee, and rubber.

Q14. Explain how irrigation has transformed the agricultural landscape of India. Give five examples of how cropping patterns have changed.

Q15. Explain the various revolutions in Indian agriculture — Green, White, Blue, Yellow, and Gene Revolutions — and their significance.

Q16. Describe the major food crops of India — rice, wheat, millets, maize, and pulses — with their geographical conditions and producing states.

Q17. Describe the major cash crops of India — sugarcane, cotton, jute, and oilseeds — with their geographical conditions, producing states, and industrial significance.

Q18. "Agriculture in India faces multiple challenges even after decades of independence." Discuss five major challenges and suggest solutions.

Q19. Explain the role of the government in developing Indian agriculture through land reforms, minimum support price, and institutional support. Give five examples.

Q20. Compare and contrast rice and wheat farming in India across five parameters — climate, rainfall, soil, season, and major producing states.

Q21. Describe the contribution of beverage crops — tea and coffee — to India's economy. Explain their geographical conditions, distribution, and export significance.

Q22. Explain how commercial farming has transformed certain regions of India. Discuss with reference to Punjab, Haryana, and Maharashtra.

Q23. "Rice is grown in three types — Aus, Aman, and Boro — in different seasons." Explain the distribution, season, and significance of rice cultivation across different regions of India.

Q24. Explain how India has moved from food scarcity to food surplus after independence. What role did the Green Revolution, irrigation, and government policy play?

Q25. Describe the distribution and significance of oilseed crops in India — groundnut, mustard, soyabean, and sunflower. Where are they mainly grown and why?

Q26. Explain the relationship between agriculture and industry in India. How do raw materials from agriculture support major industries?